scifi-RNA-seq: pre-index the cells and droplet overloading stops being a failure mode

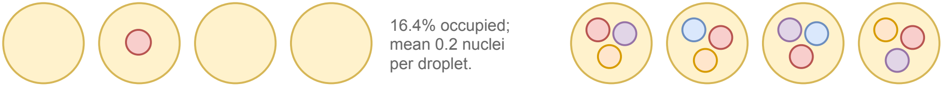
Combinatorial fluidic indexing of Datlinger et al. 2021. Poisson loading limits a droplet run only while each droplet must hold exactly one cell. Barcoding the cells first removes that requirement, so the same Chromium channel can be loaded a hundredfold harder. Colors and the 5′-to-3′ convention follow Figs. 2 and 3.

a What overloading a Chromium channel actually does, counted under a microscope

Both regimes ran without clogging the device. The droplets are drawn schematically; the counts are the measured means. In the separate yield run, 383,000 loaded nuclei returned 151,788 transcriptomes from one channel.

maximum recommended load – 15,300 nuclei per channel

100× overloaded – 1.53 million nuclei per channel



16.4% occupied; mean 0.2 nuclei per droplet.

95.5% fill; mean 9.6 nuclei per droplet.

In an ordinary droplet run every one of the right-hand droplets is a discarded doublet. Pre-indexing is what converts them into usable cells.

b Two rounds, two places: the plate writes round1 into the cell, the droplet ligates round2 on top

Round1 is shared by every cell from one well, round2 by every cell in one droplet. Neither alone identifies a cell.

1. Pre-index in a plate. Permeabilized cells or nuclei get a well-specific barcoded oligo-dT primer and reverse transcription happens in situ, on one 384-well plate. The primer also carries the UMI and the sequencing primer site.

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2. Pool and overload. Cells from every well are mixed and loaded so that most droplets receive several of them. Because the cells already differ by round1, a shared droplet is no longer a collision.

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3. Ligate round2 in the droplet. A Chromium gel bead delivers the droplet-specific barcode, joined to the 5′-phosphate of the RT primer by a thermostable ligase over a bridge oligonucleotide.

round1 × round2 **identifies one cell.**

Chromium supplies 737,280 round2 barcodes, so modeling put 1 million resolvable transcriptomes within reach of only 96 round1 wells – a modeled bound, not a measurement.

c The molecule, in the notation of Figs. 2 and 3: round2 prepends to round1 by the same splint ligation split-pool uses

Read the molecule against Fig. 2: identical grammar, but the second barcode is delivered by a droplet rather than by a second plate.

ligation junction, in the droplet

5′ | round2 BC (ligated) | primer site | round1 BC | UMI | dT | cDNA | i7 + library barcode | **3′**

written in the plate, before the cells ever met a droplet

added in bulk

after the emulsion is broken: template switch, amplification, tagmentation with an i7-only transposome

scifi-RNA-seq is the split-pool trick and the droplet trick composed, and the composition is what removes the Poisson tax. Split-pool writes a cell's address over successive plate rounds and never isolates anything (Fig. 2); droplet buys one container per cell and pays Poisson for it, keeping occupancy low (Fig. 3). Pre-indexing supplies the first half of the address in a plate, which leaves the droplet responsible only for the second half, so several cells per droplet becomes the operating point rather than the failure. The cost consequence: the per-channel reagent charge that dominates the droplet budget divides across 7.6× more cells, so 137 channels become 18 (Sec. 5.5).

Segments transcript / cDNA | round1 (plate) barcode | round2 (droplet) barcode | UMI | droplet / oil | handle / adapter | Illumina index | red dashed = the reading that changes